

Data Cleaning & Reporting Automation

Project Summary

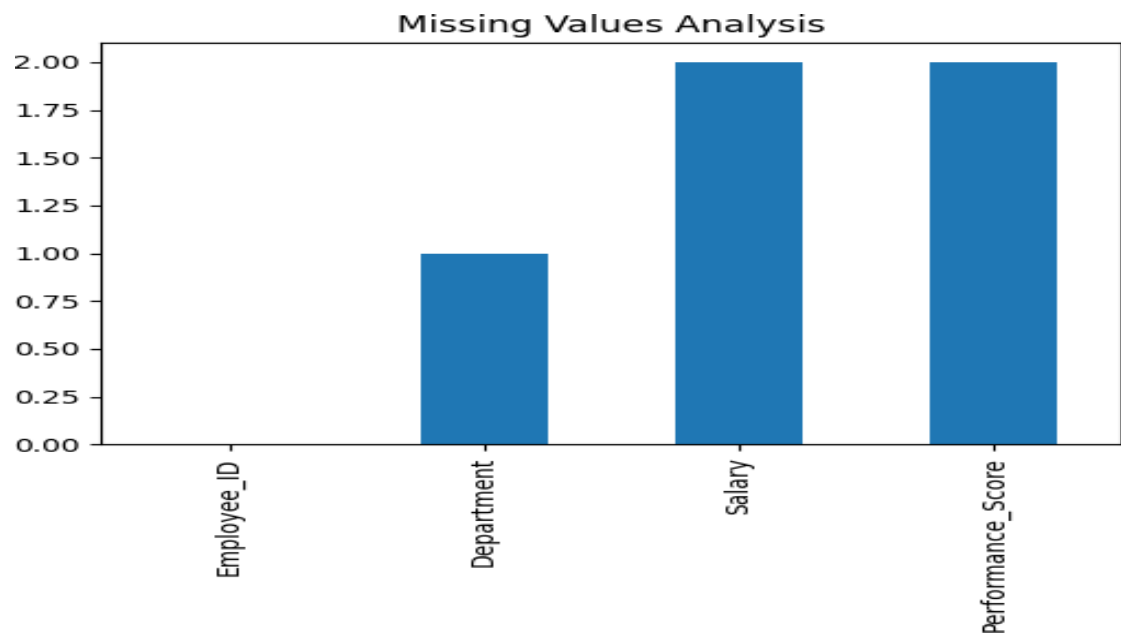
This project focuses on automating data cleaning and reporting workflows using Python. The primary objective is to identify missing values, remove duplicate records, correct inconsistent data, and generate automated reports with visual summaries for better business analysis.

The dataset contains employee-related information including department, salary, and performance scores. Several preprocessing techniques were applied to clean the data and improve data quality. The project demonstrates how automation helps organizations save time, reduce errors, and improve reporting efficiency using data preprocessing techniques.

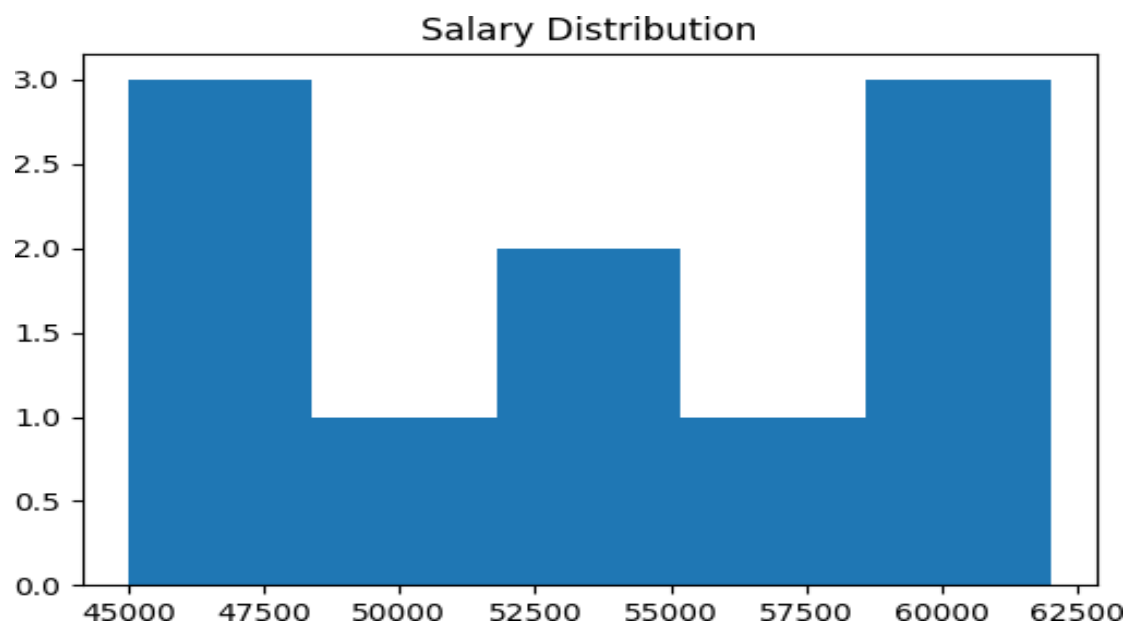
Automated visualizations and summary reports were created to analyze missing values, department distribution, salary patterns, and employee performance trends.

Task	Automation Technique
Missing Values Handling	Mean/Default Value Imputation
Duplicate Removal	drop_duplicates()
Visualization	Matplotlib Graphs
Reporting	Automated PDF Report

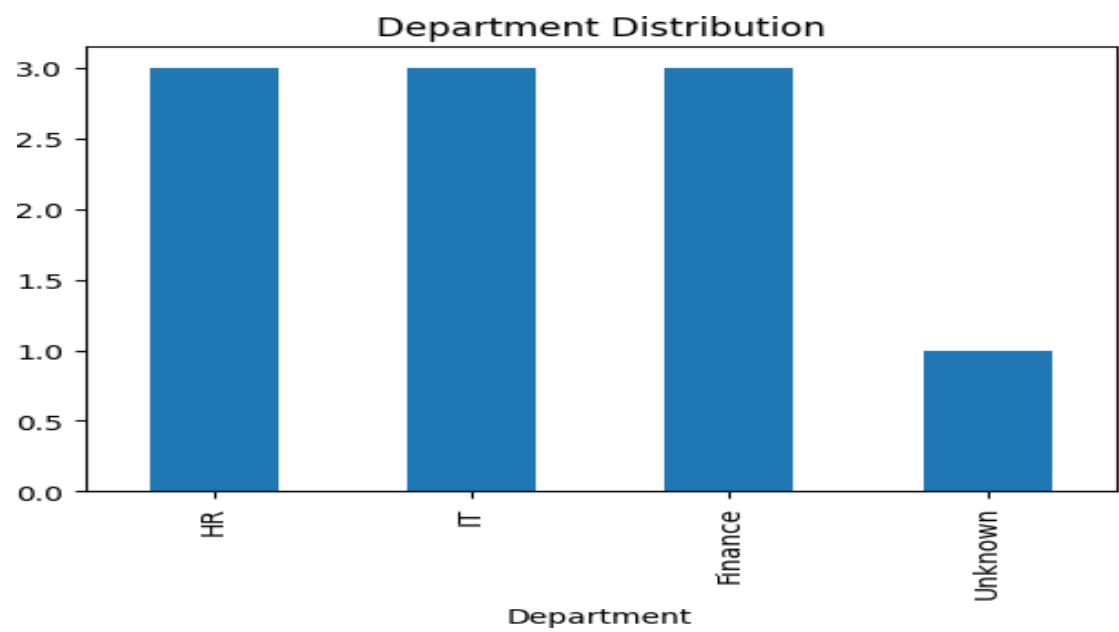
Missing Values Analysis



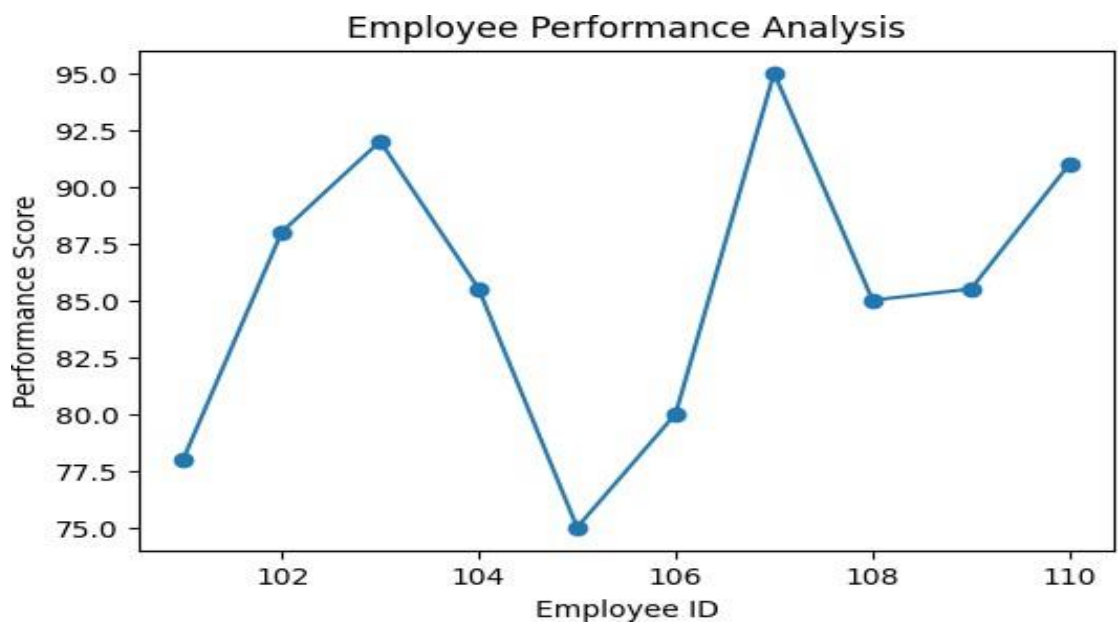
Salary Distribution



Department Distribution



Employee Performance Analysis



Conclusion

The Data Cleaning & Reporting Automation project successfully demonstrated how Python can automate data preprocessing and reporting tasks. Missing values, duplicate records, and inconsistent entries were identified and cleaned effectively using automated techniques.

The generated visualizations and summary reports provided meaningful insights into employee performance, salary distribution, and departmental analysis. This project highlights the importance of automation in improving reporting efficiency, data quality, and business decision-making processes.

Through this project, key concepts such as data preprocessing, automation workflows, reporting efficiency, and visual analytics were successfully implemented and analyzed.